Requirements Analysis Document

Kasama Support Hub Management System

Project Name: Kasama Support Hub Management System
Client: Kasama Support Hub Department
Developer: Mike Laurence A. Dela Cruz
Date: December 16, 2025

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1. Executive Summary

This document outlines the requirements for a web-based management system for the Kasama Support Hub Department. The system consists of two integrated systems: **(1) Student Concern Management System with Appointment Scheduling subsystem, and (2) Student Concern Analytics and Reporting System.** These systems address the challenges faced by Coach Hannah in managing approximately 500 students, tracking concerns submitted through various channels, and generating data-driven insights for better decision-making.

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2. Stakeholders

- **Primary Users:** Students (500+), Coach Hannah
- **System Administrator:** Mike Laurence A. Dela Cruz

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3. Functional Requirements

3.1 System 1: Student Concern Management System

FR1: User Authentication and Login

Description: The system shall provide secure login functionality for students and coaches.

Acceptance Criteria:

- Students can log in using their Student ID and password
- Coaches can log in using their staff credentials
- System validates credentials against the database within 3 seconds
- Failed login attempts are limited to 5 tries before temporary lockout

Priority: High
Rationale: Security is important since users must be authenticated to access the system.

FR2: Submit Concern

Description: Students shall be able to submit concerns through a web form with categorization.

Acceptance Criteria:

- Form includes fields for: Name, Student ID, Email, Category, Subject, Description, Urgency Level
- Students can select from 5 categories: Academic, Personal, Financial, Mental Health, Others.
- Students can optionally submit concerns anonymously
- System validates all required fields before submission
- System generates a unique Concern ID (format: C#####) upon successful submission
- Confirmation message displays the Concern ID and submission timestamp

Priority: High
Rationale: This is the core function identified in the elicitation phase. Students currently struggle with only two-way communication channels (Messenger and walk-ins).

FR3: View Frequently Asked Questions (FAQ)

Description: Students shall be able to view and search frequently asked questions before submitting concerns.

Acceptance Criteria:

- FAQ page displays questions organized by category
- Search function allows keyword search across all FAQs
- Each FAQ shows question and detailed answer
- System tracks FAQ view count for analytics purposes

Priority: Medium
Rationale: Coach Hannah mentioned "repeating questions" as a problem. FAQ reduces repetitive concerns and coach workload.

FR4: Track Concern Status

Description: Students shall be able to view the status of their submitted concerns.

Acceptance Criteria:

- Students can view a list of all their submitted concerns
- Each concern displays: Concern ID, Subject, Category, Status, Submission Date
- Status options: Pending, In Progress, Resolved
- Status is color-coded for visual clarity (yellow=pending, blue=in progress, green=resolved)
- Students can click on a concern to view full details and coach responses

Priority: High
Rationale: Addresses the problem of students not knowing the status of their concerns and having to follow up manually.

FR5: Follow-up on Concern

Description: Students shall be able to add follow-up comments or additional information to existing concerns.

Acceptance Criteria:

- Students can access their concern detail page
- A comment/reply section allows students to add text updates
- Follow-up comments are timestamped
- Coach receives notification when a follow-up is added
- Concern status can be reopened if marked as resolved but student adds follow-up

Priority: Medium
Rationale: Supports ongoing communication and prevents the need for creating duplicate concerns.

FR6: Notification System

Description: The system shall send notifications to students and coach Hannah about concern updates.

Acceptance Criteria:

- Students receive email notification when concern status changes
- Students receive notification when coach responds to their concern
- Coaches receive notification when new concern is submitted
- Coaches receive notification when student adds follow-up
- Notifications include concern ID, subject, and direct link to concern

Priority: Medium
Rationale: Keeps both students and coach Hannah informed without requiring constant system checking.

FR7: Coach View All Concerns

Description: Coach Hannah shall be able to view all submitted concerns in a centralized dashboard.

Acceptance Criteria:

- Dashboard displays all concerns with filtering options (by status, category, date, urgency)
- Concerns are sortable by submission date, urgency, or status
- Search function allows coaches to find specific concerns by Concern ID, student name, or keyword
- Dashboard shows count of pending, in-progress, and resolved concerns
- Unread/new concerns are visually highlighted

Priority: High
Rationale: Coach Hannah needs a centralized view to manage 500 students efficiently instead of Messenger conversations.

FR8: Coach Respond to Concern

Description: Coach Hannah shall be able to respond to student concerns through the system.

Acceptance Criteria:

- Coach can open concern detail page
- Text area allows coach to type response
- Coach can update concern status when responding
- Response is timestamped and attributed to the coach
- Student receives notification upon coach response

Priority: High
Rationale: Enables direct communication within the system, replacing Messenger conversations.

FR9: Update Concern Status

Description: Coach Hannah shall be able to update the status of concerns as they progress.

Acceptance Criteria:

- Coach can change status from Pending → In Progress → Resolved
- Status change is logged with timestamp
- Student is notified of status change
- Coach can add resolution notes when marking as resolved

Priority: High
Rationale: Provides transparency and helps track progress on addressing concerns.

3.2 Subsystem: Appointment Scheduling System

FR10: View Available Time Slots

Description: Students shall be able to view Coach Hannah's available appointment time slots.

Acceptance Criteria:

- Calendar view displays available time slots for the current week and next week
- Each time slot shows date, time, and availability status (available, booked, blocked)
- Calendar is updated in real-time when appointments are booked

Priority: Medium
Rationale: Addresses the problem of walk-in interruptions and allows students to plan ahead.

FR11: Book Appointment

Description: Students shall be able to book face-to-face appointments with coach Coach Hannah..

Acceptance Criteria:

- Students can select an available time slot from the calendar
- Booking form includes: Date, Time, Purpose/Reason
- Students can optionally link appointment to an existing concern
- System confirms booking and generates Appointment ID
- Both student and coach receive booking confirmation

Priority: Medium
Rationale: Replaces walk-in visits with scheduled appointments, improving time management for coaches.

FR12: Manage Coach Calendar

Description: Coach Hannah shall be able to manage their appointment calendar and availability.

Acceptance Criteria:

- Coach can set available time slots for appointments
- Coach can block time slots when unavailable
- Coach can view all booked appointments
- Coach can cancel or reschedule appointments with notification to student
- Calendar integrates with concern tracking (shows which concerns have linked appointments)

Priority: Medium
Rationale: Gives coach Hannah control over their schedule and prevents overbooking.

3.3 System 2: Student Concern Analytics and Reporting System

FR13: View Analytics Dashboard

Description: Coach Hannah shall be able to view a visual analytics dashboard with key metrics and insights.

Acceptance Criteria:

- Dashboard displays total concerns submitted (daily, weekly, monthly)
- Bar chart shows concern breakdown by category
- Displays average response time and resolution time
- Shows count of pending, in-progress, and resolved concerns
- Highlights top 3 most common concern types

Priority: High
Rationale: Coach Hannah needs data insights to manage 500 students effectively and identify patterns.

FR14: Identify Students Needing Attention

Description: The system shall flag students who may need proactive support based on their concern patterns.

Acceptance Criteria:

- System identifies students with 3+ unresolved concerns
- System flags students who submit multiple concerns within a short time period (1 week)
- System highlights students with recurring concerns of the same type
- Dashboard displays "Students Needing Attention" list with student names and concern summaries
- Coach can click on student to view full concern history

Priority: Medium
Rationale: Enables proactive intervention before students reach crisis point, addressing coaches' need to monitor student wellbeing.

FR15: Generate Reports

Description: The system shall generate automated reports summarizing concern data and trends.

Acceptance Criteria:

- System generates daily summary report (concerns received today, resolved today)
- System generates weekly report (trends, patterns, top concerns)
- System generates monthly comprehensive report (statistics, student risk analysis, response time metrics)
- Reports can be exported as PDF or Excel format
- Reports include charts and visualizations

Priority: Medium
Rationale: Coach Hannah mentioned the need for reports and monitoring. Automated reports save time and provide documentation.

FR16: Track Response and Resolution Time

Description: The system shall automatically calculate and track time metrics for concern handling.

Acceptance Criteria:

- System calculates time from concern submission to first coach response
- System calculates time from concern submission to resolution
- Dashboard displays average response time and resolution time
- System identifies concerns that exceed expected response time (e.g., >24 hours)
- Metrics are broken down by concern category

Priority: Low
Rationale: Provides performance metrics to help improve service quality and identify bottlenecks.

FR17: Trend Analysis

Description: The system shall analyze concern data to identify patterns and trends.

Acceptance Criteria:

- System compares current week data to previous week data
- System identifies increases or decreases in specific concern categories
- Dashboard displays alerts for significant spikes (e.g., "Mental Health concerns increased 40% this week")
- System shows seasonal patterns (e.g., more concerns during exam week, enrollment period)

Priority: Low
Rationale: Helps coach Hannah anticipate busy periods and prepare appropriate resources proactively.

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4. Non-Functional Requirements

NFR1: Performance

Description: The system shall respond to user actions within acceptable time limits.

Acceptance Criteria:

- Page load time: Maximum 3 seconds on standard internet connection
- Form submission processing: Maximum 2 seconds
- Search results: Maximum 2 seconds
- Report generation: Maximum 10 seconds for monthly reports

Priority: High
Rationale: Students and coach Hannah need quick access to information since slow systems discourage usage.

NFR2: Security

Description: The system shall protect user data and ensure privacy.

Acceptance Criteria:

- All passwords are encrypted using industry-standard hashing
- Student concerns are only visible to the student who submitted them and authorized coaches
- Anonymous concerns do not reveal student identity to coaches
- System uses HTTPS for all data transmission
- Session timeout after 30 minutes of inactivity

Priority: High
Rationale: Student concerns often contain sensitive personal information requiring protection.

NFR3: Accessibility

Description: The system shall be accessible to users with diverse needs and devices.

Acceptance Criteria:

- Minimum font size of 14px for body text
- Color contrast ratio meets WCAG 2.1 Level AA standards (minimum 4.5:1)
- All form fields have clear, descriptive labels
- Keyboard navigation is fully supported
- Error messages are clear and provide actionable guidance

Priority: Medium
Rationale: Ensures all students, including those with visual impairments or disabilities, can use the system.

NFR4: Compatibility

Description: The system shall work across different browsers and devices.

Acceptance Criteria:

- Compatible with Chrome, Firefox, Safari, Edge (latest 2 versions)
- Responsive design adapts to mobile phones (320px width minimum)
- Responsive design adapts to tablets (768px width)
- Responsive design adapts to desktop (1024px width and above)
- Core functionality works without JavaScript (graceful degradation)

Priority: High
Rationale: Students primarily use mobile phones; the system must be mobile-friendly.

NFR5: Availability

Description: The system shall be available for use at all times.

Acceptance Criteria:

- System uptime of 99% or higher
- Scheduled maintenance windows communicated 48 hours in advance
- System accessible 24/7 as students may need to submit concerns outside office hours

Priority: Medium
Rationale: Students may experience concerns or need support at any time, including evenings and weekends.

NFR6: Usability

Description: The system shall be easy to use without extensive training.

Acceptance Criteria:

- New users can submit a concern within 2 minutes of first login
- Navigation menu is consistently placed on all pages
- Error messages explain what went wrong and how to fix it
- Success messages confirm actions clearly
- Interface uses plain language, avoiding technical jargon

Priority: High
Rationale: Complex systems discourage usage; simplicity encourages students to use the system instead of Messenger.

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5. Requirements Priority Summary

High Priority (Must Have):

- FR1: User Authentication
- FR2: Submit Concern
- FR4: Track Concern Status
- FR7: Coach View All Concerns
- FR8: Coach Respond to Concern
- FR9: Update Concern Status
- FR14: View Analytics Dashboard
- NFR1: Performance
- NFR2: Security
- NFR4: Compatibility
- NFR6: Usability

Medium Priority (Should Have):

- FR3: View FAQ
- FR5: Follow-up on Concern
- FR6: Notification System
- FR10: View Available Time Slots
- FR11: Book Appointment
- FR13: Manage Coach Calendar
- FR15: Identify Students Needing Attention
- FR16: Generate Reports
- NFR3: Accessibility
- NFR5: Availability

Low Priority:

- FR17: Track Response and Resolution Time
- FR18: Trend Analysis

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6. Traceability to Elicitation

Interview with Coach Hannah:

- **"Messenger, walk-ins, forms"** → FR2 (centralized submission system)
- **"500 students"** → FR14, FR15 (analytics to manage large volume)
- **"Repeating questions"** → FR3 (FAQ system)
- **"Need reports"** → FR16 (automated report generation)
- **"Too many students to handle all alone"** → FR7 (centralized dashboard)
- **"Sometimes na-miss or na-forget"** → FR4, FR6 (tracking and notifications)

Observation (Expected findings):

- Students waiting → FR10, FR11 (appointment scheduling)
- Walk-in interruptions → FR13 (calendar management)
- Repetitive questions → FR3 (FAQ)
- Lost concerns → FR4, FR9 (status tracking)

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7. Assumptions and Constraints

Assumptions:

- Students have access to internet-connected devices (computers or smartphones)
- Students have valid student ID credentials for login
- Coach Hannah has administrative access to manage the system
- School IT infrastructure supports hosting a web-based application

Constraints:

- Development time: Approximately 2 months
- Technology stack: PHP, MySQL, HTML, CSS, JavaScript (as specified in project requirements)
- No budget for third-party premium services or APIs
- System must work without requiring app installation (web-based only)

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8. Acceptance Testing Overview

Each requirement will be validated through:

- **Unit Testing:** Individual functions tested in isolation
- **Integration Testing:** System components tested together
- **User Acceptance Testing (UAT):** Coach Hannah and sample students test the system
- **Performance Testing:** Load testing to ensure system handles 500+ concurrent users

Success criteria:

- All High Priority requirements fully functional
- At least 80% of Medium Priority requirements implemented
- System passes all acceptance criteria defined for each requirement
- Coach Hannah approves system as meeting needs identified in elicitation

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9. Revision History

Version	Date	Author	Changes
1.0	December 16, 2025	Mike Laurence A. Dela Cruz	Initial requirements analysis document

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Approved By:

Approval Date: